Chapter 1 : PAIR

Before Vector, learn Pair.

What is Pair?

Stores 2 values together.

Think:

```
name = "Rahul"
age = 21
```

Instead of separate variables:

```
pair<string,int> p;
```

Stores:

```
("Rahul",21)
```

Declaration

```
pair<int,int> p;
```

Initialization

```
pair<int,int> p = {10,20};
```

or

```
pair<int,int> p(10,20);
```

Access Values

```
cout << p.first;
cout << p.second;
```

Example:

```
pair<int,int> p = {5,10};

cout << p.first;   // 5
cout << p.second;  // 10
```

Nested Pair

```
pair<int,pair<int,int>> p;
```

Example:

```
pair<int,pair<int,int>> p;

p = {1,{2,3}};
```

Access:

```
cout << p.first;
cout << p.second.first;
cout << p.second.second;
```

Output:

```
1 2 3
```

Array of Pairs

```
pair<int,int> arr[3];
```

Example:

```
arr[0]={1,2};
arr[1]={3,4};
arr[2]={5,6};
```

Print:

```
for(int i=0;i<3;i++)
{
    cout<<arr[i].first<<" "
        <<arr[i].second<<endl;
}
```

Pair Functions

Pair has very few functions.

---

make_pair()

Creates pair.

```
auto p = make_pair(10,20);
```

Equivalent:

```
pair<int,int> p={10,20};
```

---

swap()

Swap two pairs.

```
pair<int,int> p1={1,2};
pair<int,int> p2={3,4};

swap(p1,p2);
```

Result:

```
p1 = {3,4}
p2 = {1,2}
```

Comparison of Pairs

Very important for sorting.

Equal

```
p1 == p2
```

Not Equal

```
p1 != p2
```

Less Than

```
p1 < p2
```

Pair compares:

1. first
2. second

Example:

```
(1,5) < (2,3)
```

Because:

```
1 < 2
```

True.

Example:

```
(2,3) < (2,7)
```

Compare second:

```
3 < 7
```

True.

Time Complexity

All operations:

```
O(1)
```

Interview Uses

Coordinates

```
(x,y)
```

Graph

```
(node,weight)
```

Sorting

```
(student_marks,id)
```

Priority Queue

```
(distance,node)
```

Used in Dijkstra.

Practice Questions

Easy

Q1

Create pair:

```
(10,20)
```

Print both values.

Q2

Take input:

```
a b
```

Store in pair.

Print:

```
first second
```

Q3

Swap two pairs.

Input:

```
1 2
3 4
```

Output:

```
3 4
1 2
```

Medium

Q4

Store coordinates:

```
(1,2)
(3,4)
(5,6)
```

in array of pairs and print.

Q5

Input N pairs.

Sort them.

Example:

```
4

2 5
1 3
2 1
1 1
```

Output:

```
1 1
1 3
2 1
2 5
```

(Hint: use `sort()`)

Revision Sheet

Declaration

```
pair<int,int> p;
```

Initialize

```
pair<int,int> p={1,2};
```

Access

```
p.first
p.second
```

Create

```
make_pair()
```

Swap

```
swap()
```

Compare

```
==
!=
<
>
<=
>=
```

Complexity

```
O(1)
```

Your Assignment Before Next Chapter (Vector)

Solve:

1. Create and print pair.
2. Swap two pairs.
3. Array of pairs.
4. Sort vector of pairs.
5. Store student `(marks, roll_no)` and sort.